NISHANT JAIN

AI/ML & IoT Engineer • Computer Vision • Edge AI • Full Stack

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SUMMARY

AI/ML & IoT Engineer building production-grade Computer Vision and Edge AI systems with Python, YOLO, OpenCV, Flask, AWS, and NVIDIA Jetson. Patent applicant with experience owning the full delivery cycle — data pipelines, model training and optimization, REST APIs, and cloud/edge deployment — for real-world clients.

TECHNICAL SKILLS

AI / ML: Python, YOLOv5/v8, TensorFlow, PyTorch, Scikit-learn, OpenCV, Object Detection, Segmentation

Edge / IoT: NVIDIA Jetson Nano / Orin, CUDA, TensorRT concepts, ESP32, Arduino, PLC, UART, I2C, TCP/IP

Backend & Cloud: Flask, FastAPI, RESTful APIs, MySQL, SQLite, AWS EC2, Linux, Nginx, Gunicorn, Systemd, Git

GenAI: LLM API integration, Prompt Engineering, RAG concepts

EXPERIENCE

AI/ML & IoT Engineer — Synquic Labs Pvt. Ltd., Indore **Aug 2026 – Present**

- Develop AI/ML models, computer vision, and edge AI applications for product use cases.
- Build IoT solutions with microcontrollers and embedded devices, integrating hardware with cloud and backend systems.

Founding Engineer – AI & Software — Cognicity Pvt. Ltd., Indore **Sep 2025 – Aug 2026**

- Worked directly with clients to scope AI/CV solutions, then owned the full build: data pipeline, model training, REST API integration, and AWS cloud deployment.
- Designed and implemented YOLO-based detection systems, benchmarking results against client KPIs (mAP, precision, recall) to validate performance before release.
- Managed deployment infrastructure — Systemd services, Nginx reverse proxy, SSH-based access — for cloud and edge systems running in production.

Machine Learning & Computer Vision Intern — Cognicity Pvt. Ltd., Indore **Aug 2025 – Sep 2025**

- Fine-tuned YOLO and TensorFlow models using data augmentation and hyperparameter tuning; deployed ML inference pipelines on NVIDIA Jetson Nano for the team's first edge-AI client delivery.

PROJECTS

EdgeDesk — Secure Multi-Device Remote Access Platform (Patent Filed) | *Python, WebSocket, VNC, Jetson Nano, TLS/WSS*

- Designed and built a VPN-style secure remote desktop platform for encrypted screen access and file transfer across a fleet of Jetson Nano edge devices — patent application filed for the multi-device architecture.
- Built a custom WebSocket broker with encrypted file transfer and cross-device drag-and-drop, managing multiple edge devices from one dashboard with reboot-safe Linux services.

Real-Time Crowd Analytics & Stampede Detection | *Python, YOLOv8, ByteTrack, OpenCV, Jetson Nano (CUDA)*

- Built a GPU-accelerated multi-object tracking pipeline achieving sub-50 ms inference latency for real-time person counting and density/velocity-based stampede alerts; Top Finalist among 200+ teams — MPSEDC Mahakumbh Hackathon 2025.

AI-Accelerated Facial Recognition Attendance System | *Python, Flask, dlib, MySQL, AWS EC2*

- Built and deployed a full-stack attendance system on AWS EC2, reducing manual attendance-taking time by ~90% for a 3-course pilot through automated facial recognition.

AI-Powered Poultry Health Monitoring | *Python, YOLO, OpenCV, Object Detection*

- Owned the ML lifecycle end-to-end — data collection, annotation, training, and deployment — for a real farm-operator client.

EDUCATION

B.Tech + M.Tech Dual Degree — Internet of Things — DAVV, Indore **2021 – 2026**

Relevant coursework: DSA, DBMS, OS, Computer Networks, ML, Embedded Systems

AWARDS & CERTIFICATIONS

Winner — Sustainathon 2025 (NATRAX) | **Top Finalist — MPSEDC Mahakumbh Hackathon 2025**

Data Science Certification — Universal Informatics | IoT & Microcontrollers Training — RRCAT, DAE, Indore | PLC iQ-F — Mitsubishi Electric